

Farming Setup

Feature Specifications

Version 1.0

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Document Properties

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1.1 Purpose

This document specifies the Farming Setup feature of the AgriERP Web Application: the farm hierarchy (Grower › Site › Farm), crop years, the permanent Field Master and its crop-year-wise Field Cultivation records (one per Field ID + Crop Year, holding acres, irrigation, coordinates, varieties, planted year and other cultivation attributes), copying plots into a new crop year, and the lifecycle of the matching Project in D365 F&SCM. It defines the release plan, workflows, business use cases, validation rules, technical design and data dictionary for each release (R1–R4).

1.2 Scope

In scope: Settings › Farming Setup pages — Growers, Sites, Farms, Crop Years and Plot Cultivation — including the Filter panel, per-column search, the Add/Edit/View side panels (General, Varieties, Advanced), Delete rules, Add Crop Year with plot copy, Copy Plots from Crop Year, Create Project and Project Update with D365 F&SCM. Out of scope: mobile app screens (mobile reads plot, variety and project data only), map/KML drawing, bulk Excel import, and user-configurable hierarchy or fields (Future Considerations).

1.3 Audience

Folio3 project executives, product management, solution architects, functional consultants (D365 Project Management & Accounting), developers, QA engineers and the customer implementation team.

Introduction

Problem Statement

Previously, the farming setup was managed in D365 F&SCM. After moving this setup out of D365, the data was maintained directly in the AgriERP database through scripts, as no user interface was available to manage Growers, Sites, Farms, and Field information.
This created a dependency on the technical team for adding or updating farming setup data, as users could not manage it directly from the AgriERP Web Application.
Another key limitation was the lack of crop-year-wise field management. The same physical field may have different acreage, varieties, irrigation details, and other cultivation information across crop years. However, there was no proper way to maintain these changes separately while preserving the field's permanent identity and historical information.
A dedicated Farming Setup module is required in AgriERP to manage the existing farm hierarchy, maintain a permanent Field Master, and manage cultivation records for each Field + Crop Year combination. Each cultivation record must also support its respective D365 project.

Solution Overview

AgriERP introduces Farming Setup — one place to maintain the farm hierarchy and crop years, a permanent register of physical fields, the cultivation details of each field per crop year, and the D365 project for each cultivation.

Field Master and Crop Year data model (confirmed)

Each physical field has one permanent, unique Field ID in the Field Master that never changes across crop years. Everything that changes per season — Total Area, Operational Area, Irrigation Method and Sources, Coordinates, Varieties, Planted Year and other cultivation attributes — is stored in a Field Cultivation record identified by Field Master + Crop Year, which is unique — a Field Master can have only one cultivation record per crop year. Each Field Cultivation record has its own D365 project. Changing one crop year never affects another.

Field ID	Crop Year	Operational Area	Project
130	2026	150 acres	J01001
130	2027	148 acres	J02001
130	2028	146 acres	J03001

All three rows belong to the same physical Field 130 in the Field Master; their cultivation information and projects are maintained independently. How existing field and project records map to this structure is listed under Open Technical Questions.

- Hierarchy & crop years — the Growers, Sites and Farms that already exist in the database become visible and maintainable in AgriERP; new records follow the existing ID scheme; empty Crop Years can be created.
- Plot/Field register and crop-year-wise Plot Cultivation — the field keeps its permanent ID; each season gets its own General details, Varieties with planted area and optional Advanced attributes.
- Season roll-over — create a crop year with a copy of last year's cultivation records for the same fields, editing name, areas, irrigation, coordinates and varieties in one table before saving.
- Project creation — one D365 Project per cultivation record (Field + Crop Year), each with its own Name, Type, Group and Start Date; AgriProject is set to Yes automatically.
- Project update — when a cultivation record with a linked D365 project is saved, its D365-mapped changes are synchronized automatically to that record's own project. The exact field mappings are being validated with the D365 integration team.
- Fast navigation — a Filter panel (Crop Year, Grower, Site, Farm) and a search box under every table column.

Release Plan

R1–R4 are internal development phases used to plan, build and test the feature in increments. They are not client-facing releases; the client receives Farming Setup as one feature, and delivery milestones are agreed separately.

Release	Goal	In scope	Not in this release
R1	Hierarchy & Crop Year Setup	Existing Growers, Sites and Farms from the database are listed and searchable in AgriERP. Add new records (IDs follow the existing numbering scheme), edit names, delete when nothing is linked below. Create an empty Crop Year (Code + Name), edit Name, delete when it has no cultivation records.	Fields, cultivation records, copy, projects.

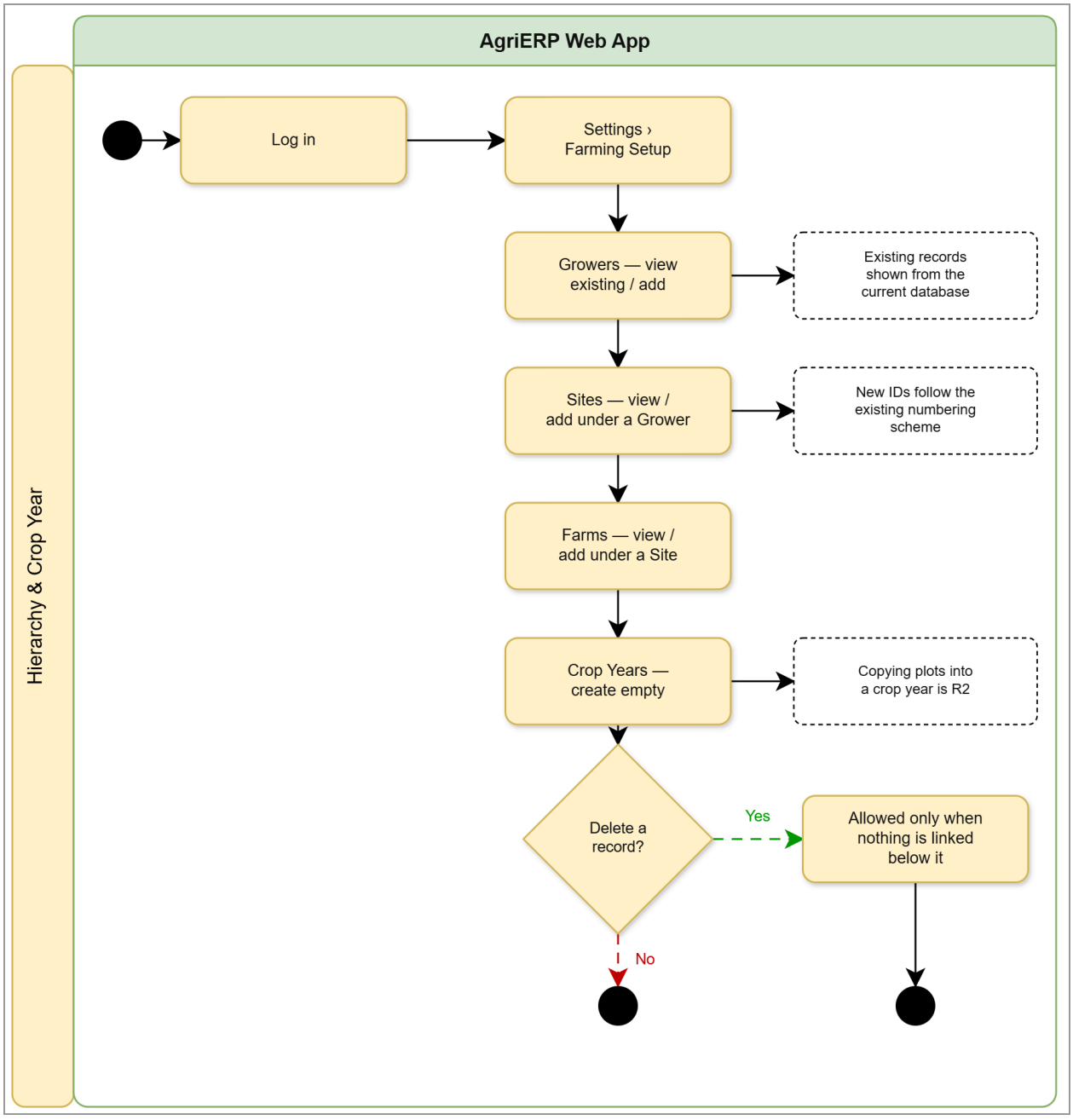
Release	Goal	In scope	Not in this release
R2	Field Master & Crop-Year-Wise Cultivation	Field Master under each Farm (existing fields shown): select an existing field or create a new one. Field Cultivation per Field Master + Crop Year: add, view, update, delete with General, Varieties and Advanced tabs; previous crop years stay accessible from the field. Create Crop Year with copy reuses the Field Master and creates new cultivation records only. Cultivation records already linked to a project are locked from day one.	Creating or updating D365 projects.
R3	D365 Project Creation	Triggered only when an authorized user clicks Create Project — saving a cultivation record never creates a project. Select one or more cultivation records without a project and Create Project. Each gets its own Project Name, Project Type, Project Group and Project Start Date. D365 creates the project with AgriProject = Yes set automatically and returns the Project No. The same lock as existing linked projects then applies. Failure status and retry.	Updating an existing project.
R4	D365 Project Update	Once a D365 project is linked to a Field Cultivation record, any updates to its D365-mapped fields must be synchronized automatically when the user saves the changes in AgriERP. Updates to fields that are not mapped to D365 remain in AgriERP only. Sync runs on Save with no separate action, goes only to the project of that Field + Crop Year, shows Updating… then Synced (last synced time) or Update failed with Retry; Retry re-sends to the same project and never creates a new one. Cultivation records with a project are editable except Plot/Field ID, Crop Year and hierarchy; Delete stays blocked.	Deleting or closing a D365 project from AgriERP. Which fields are mapped is decided by the D365 mapping review (Project Name, Total/Operational Area, Irrigation Method are To confirm) — that review decides the field list, not whether sync happens.

Summary

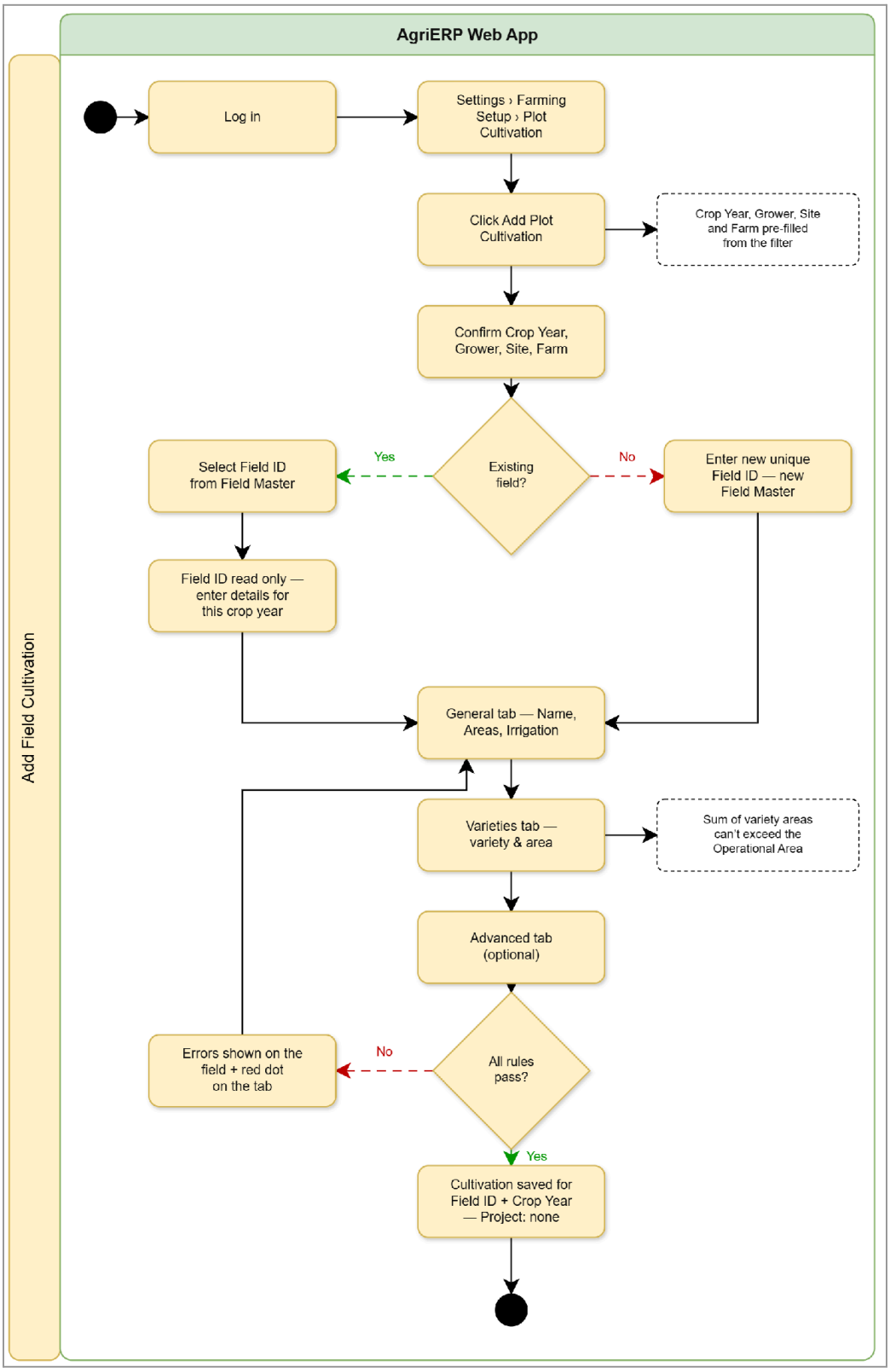
Each phase is usable on its own: R1 exposes and maintains the existing ownership structure and seasons, R2 delivers the field register with yearly cultivation records and one-step roll-over, R3 creates a D365 project for every cultivation record, and R4 keeps those projects in step with the field.

Workflow

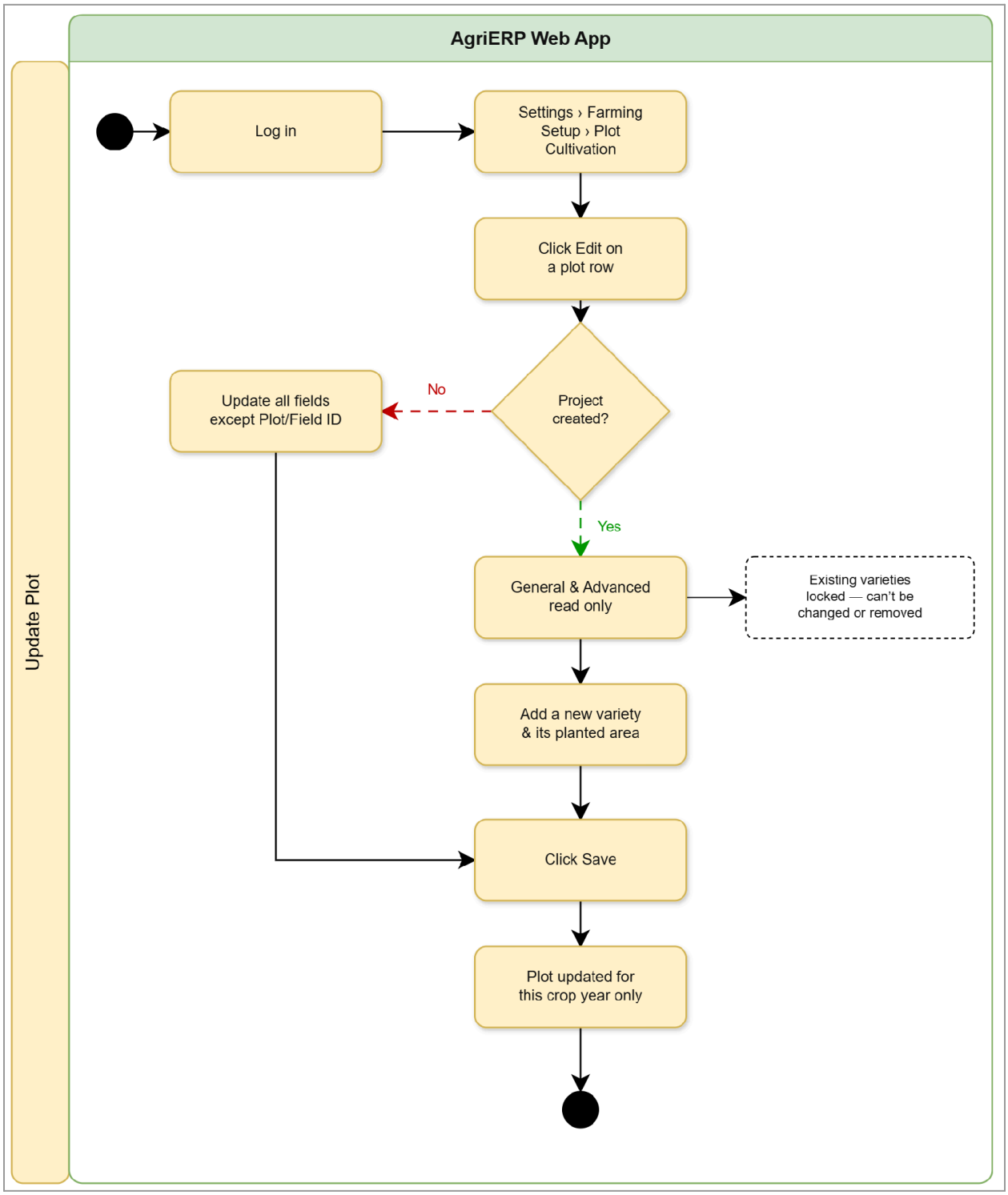
R1 · Hierarchy — Grower › Site › Farm



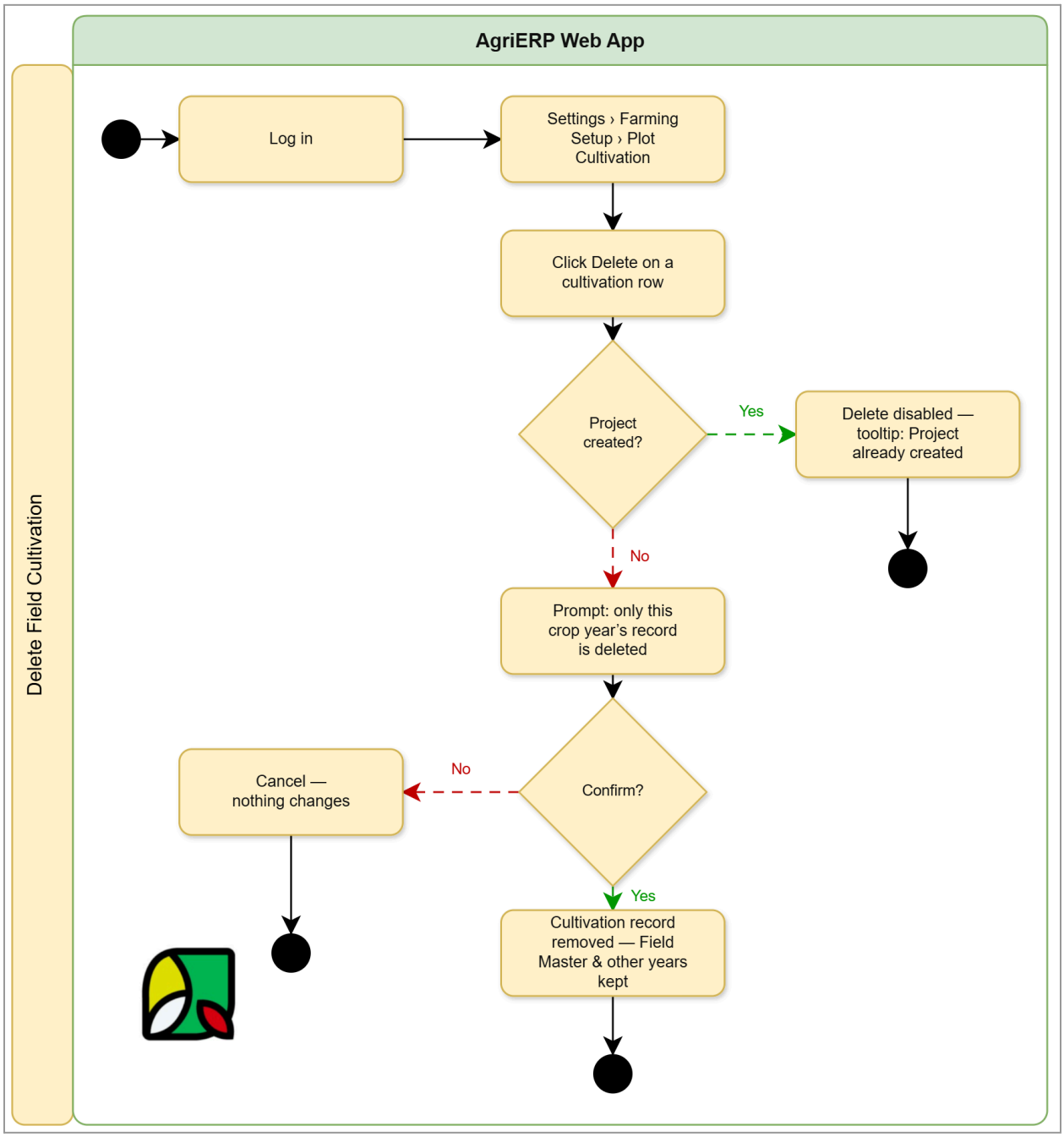
R2 · Add Field Cultivation (Existing Field / New Field)



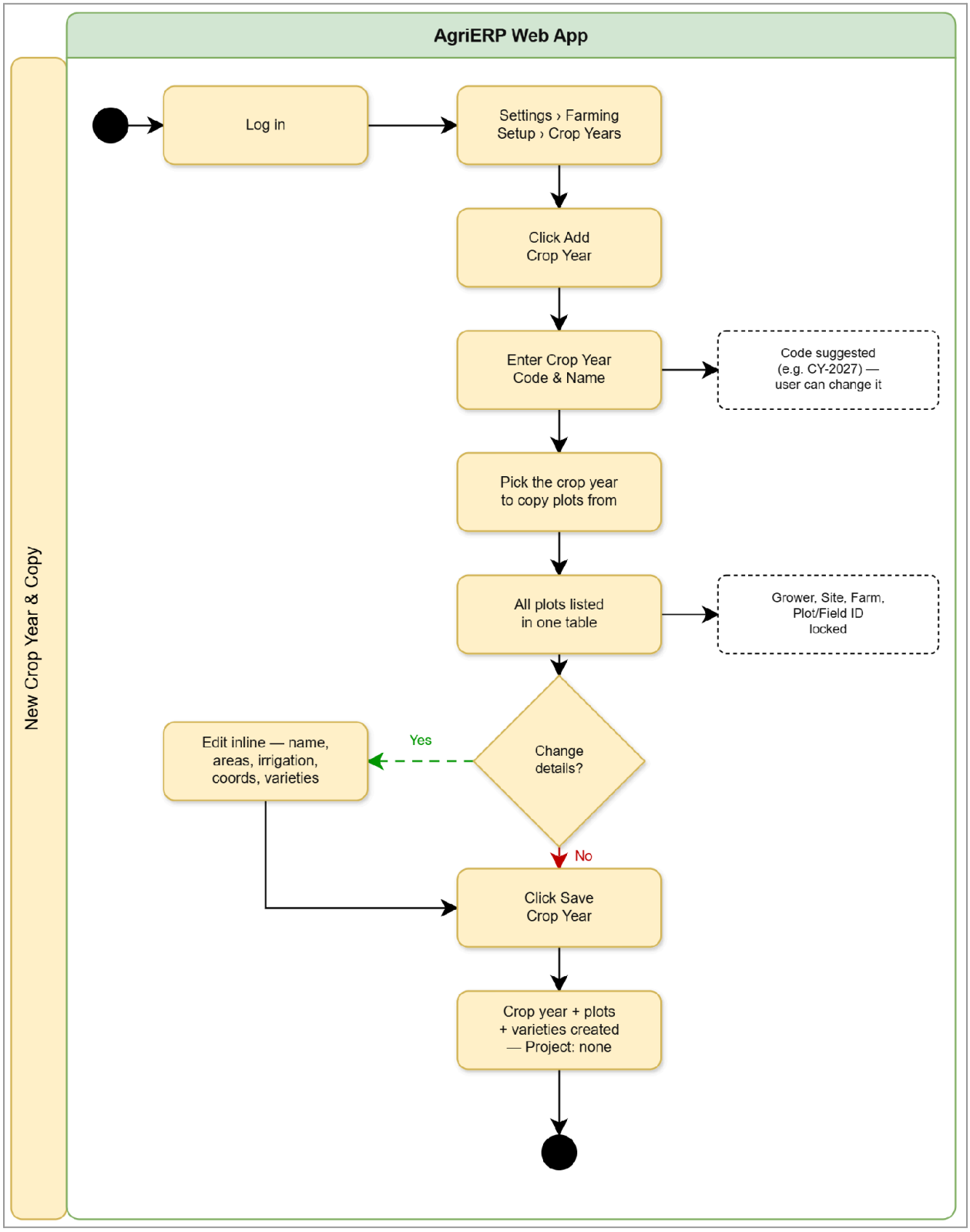
R2 · Update Plot Cultivation (lock applies to any linked project)



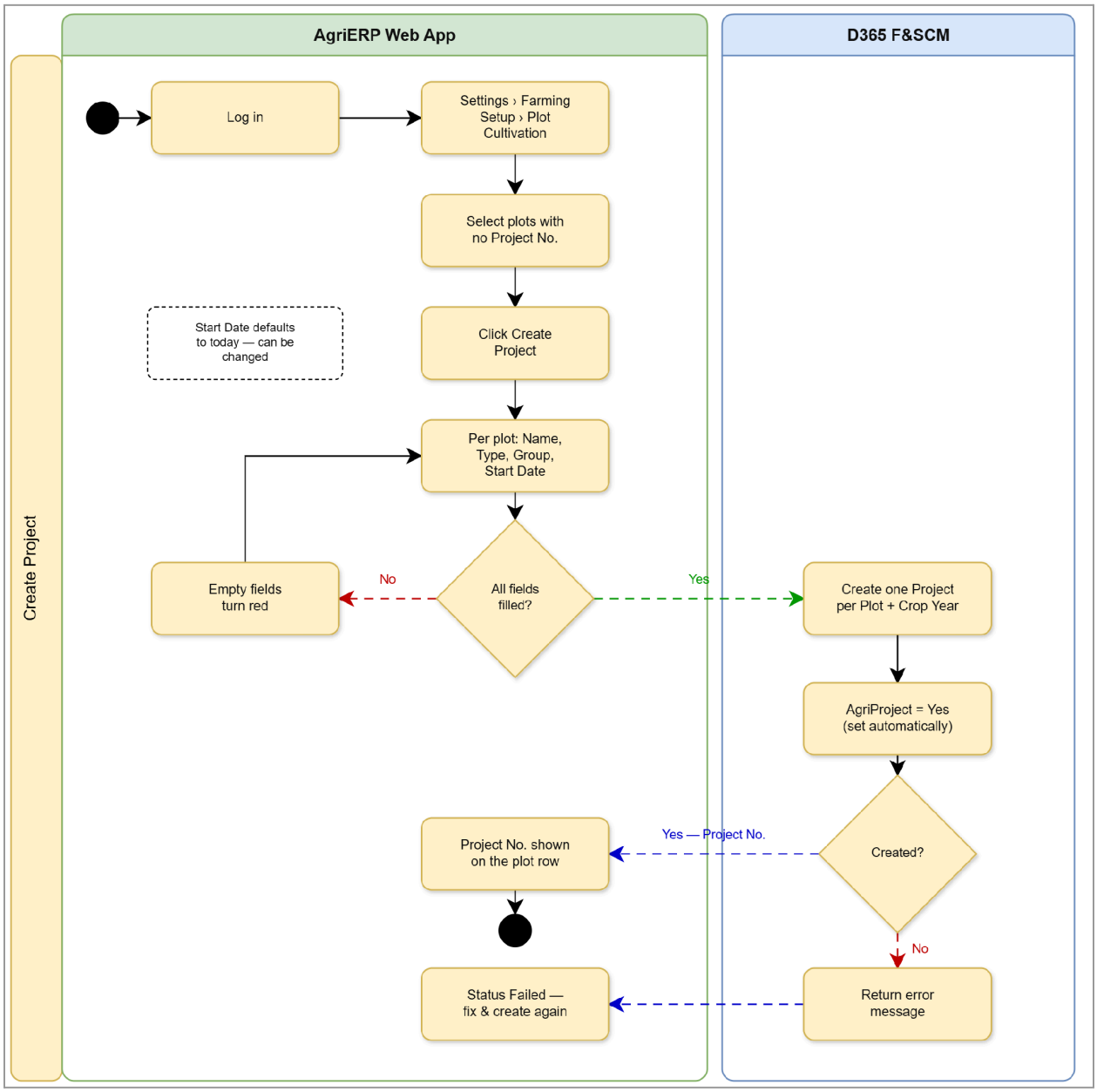
R2 · Delete Field Cultivation



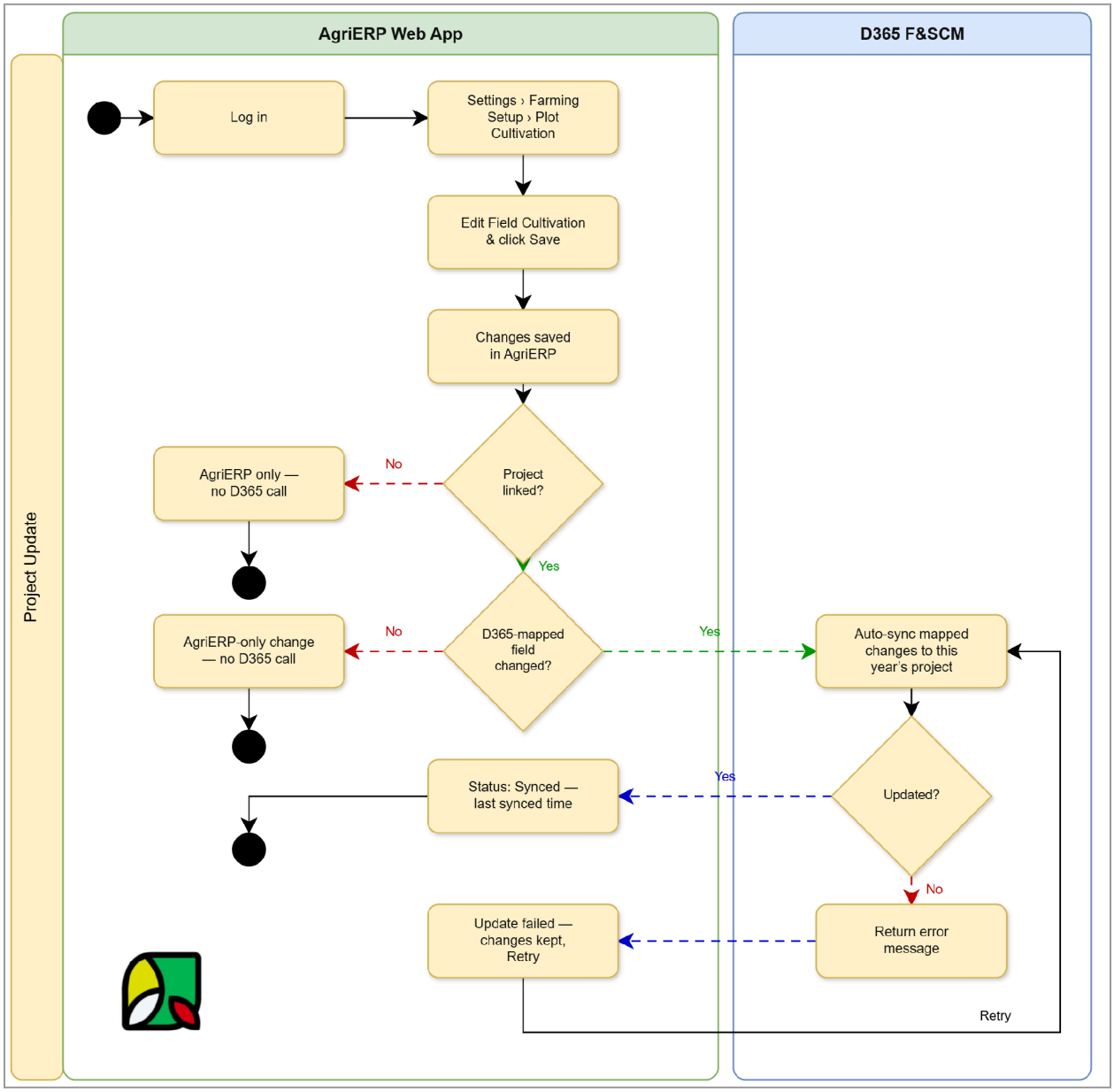
R2 · Create Crop Year with Plot Copy



R3 · Create Project in D365

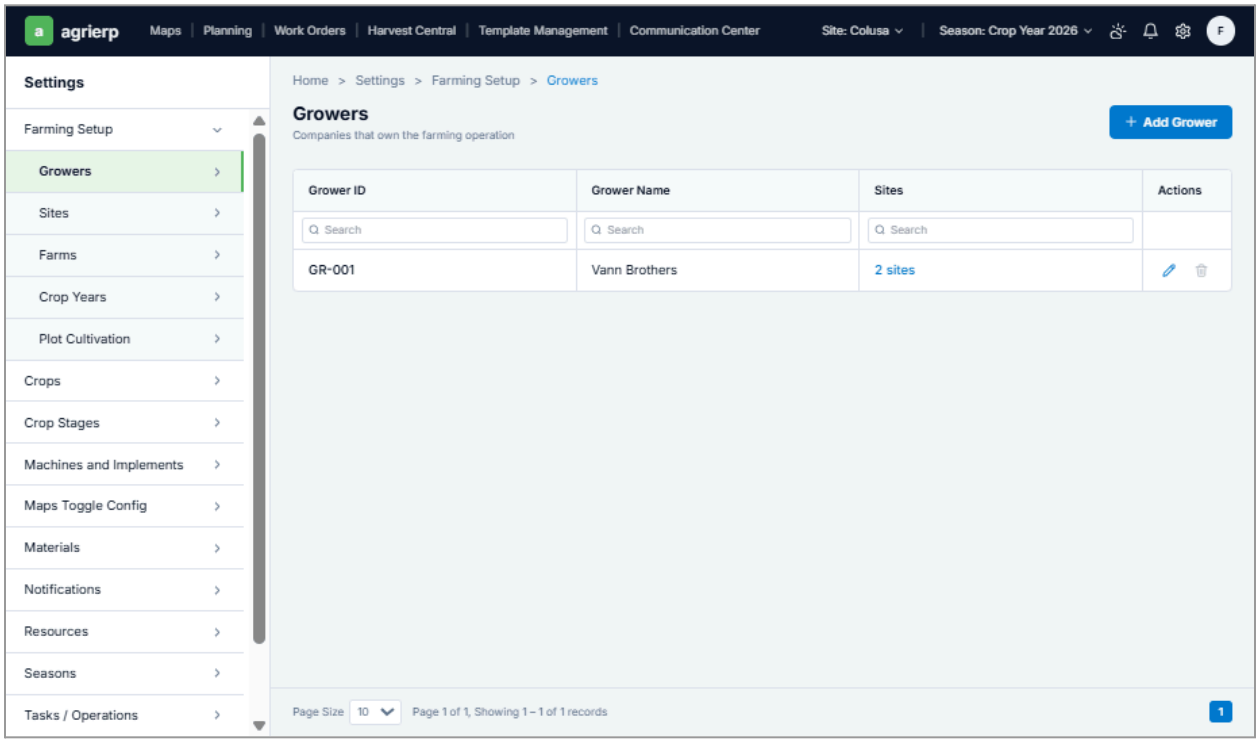


R4 · Project Update from AgriERP

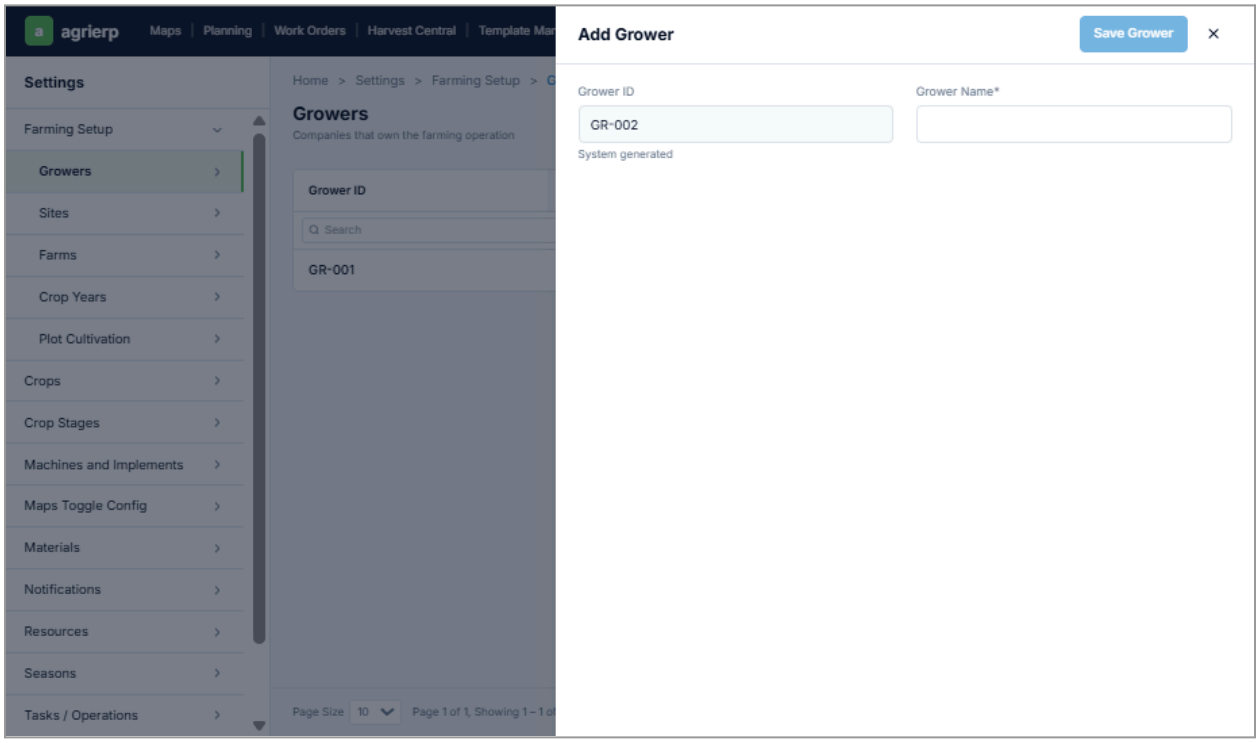


Mockups

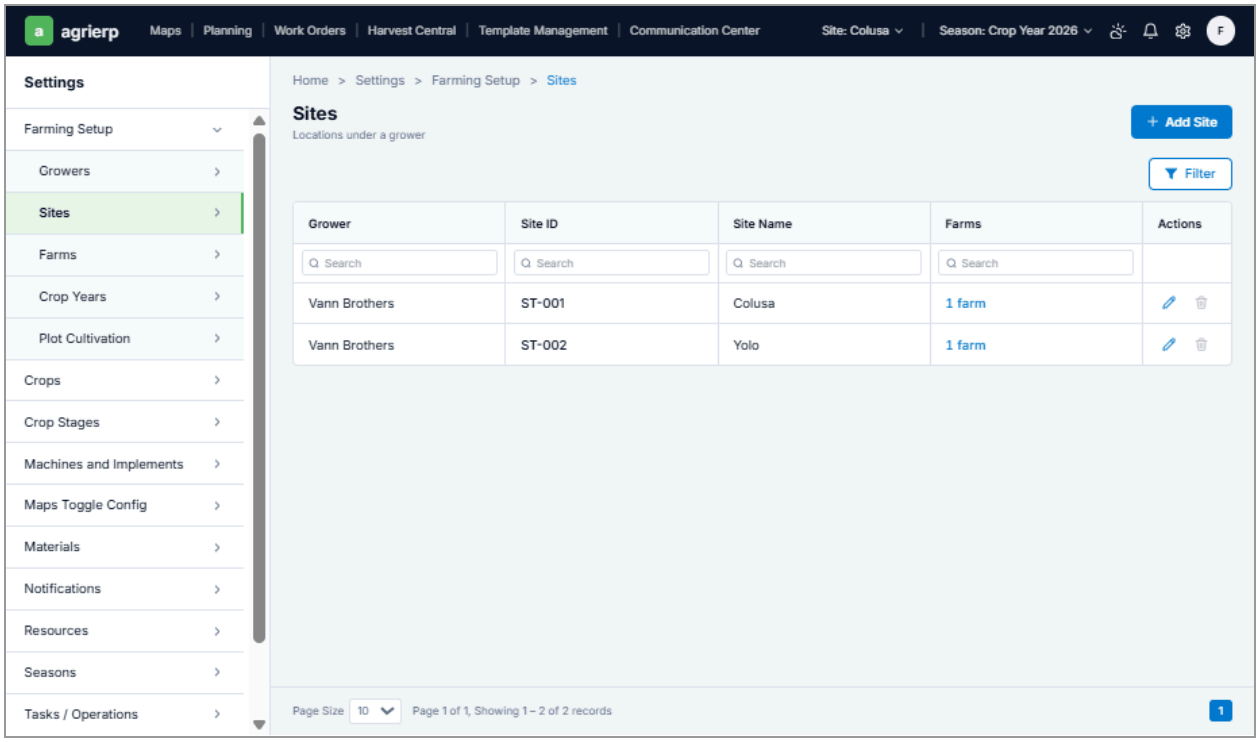
R1 · Growers



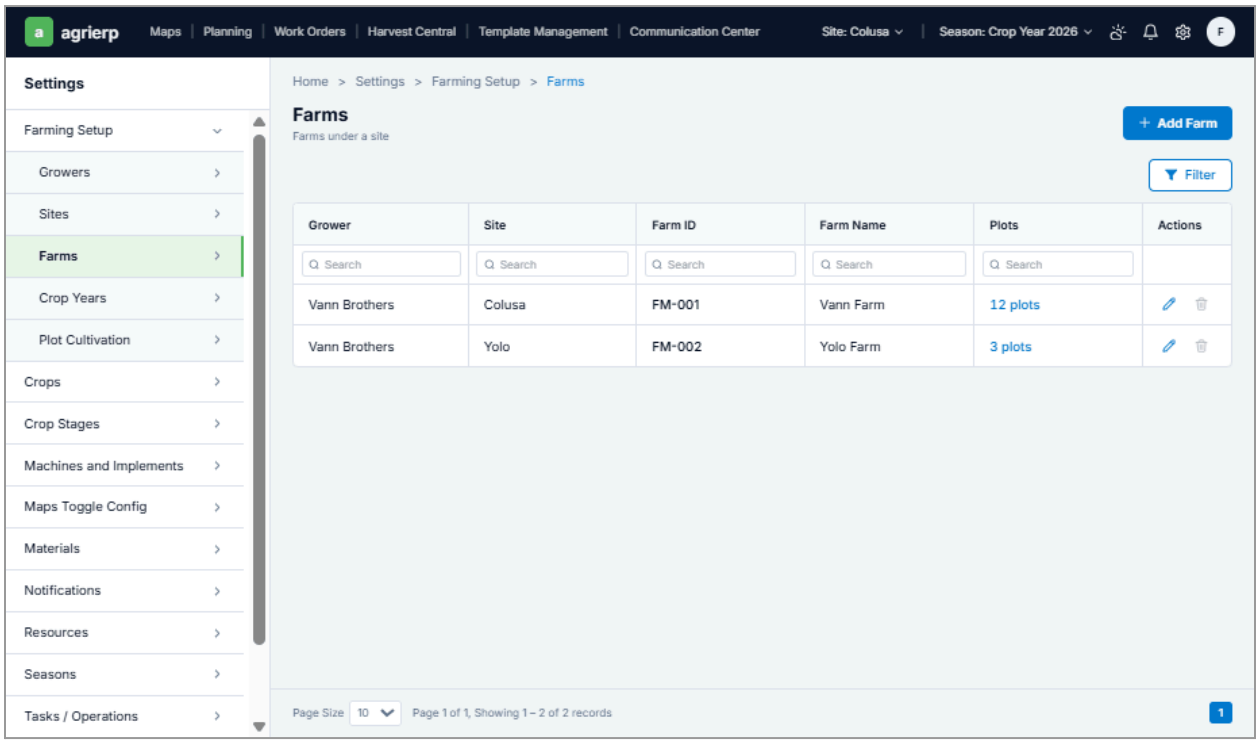
R1 · Add Grower — Grower ID system generated



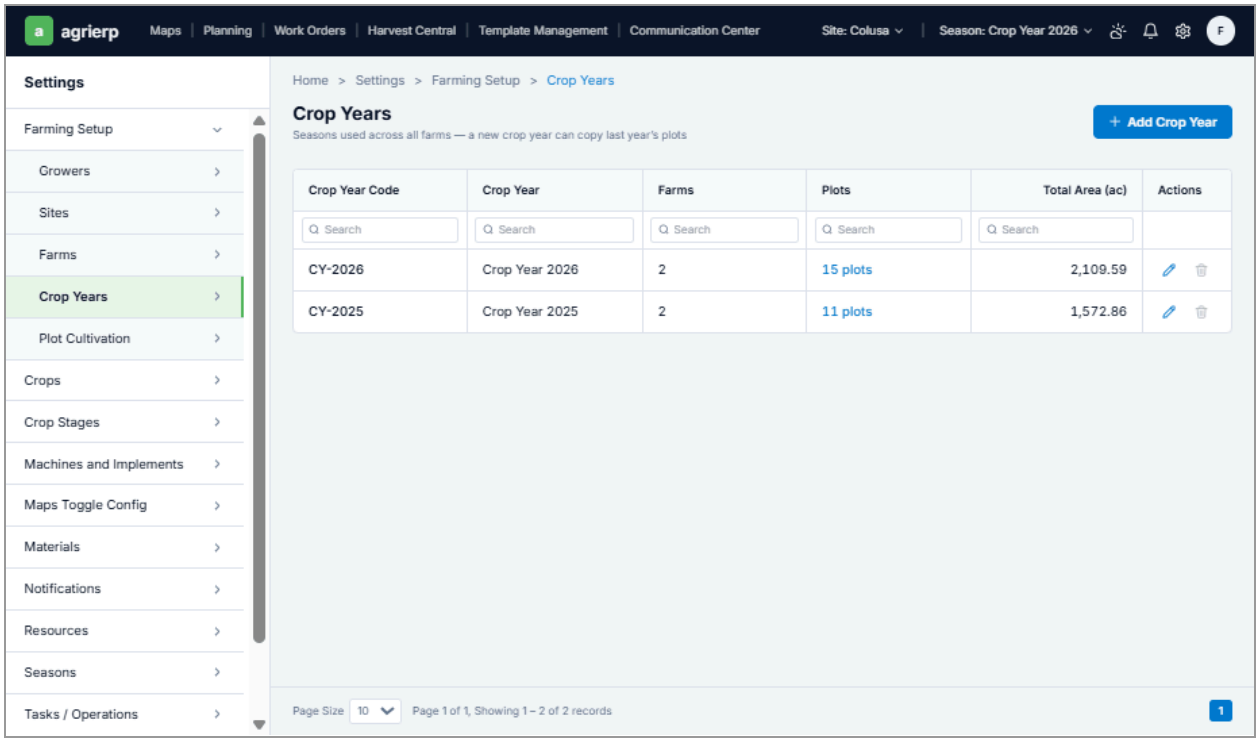
R1 · Sites



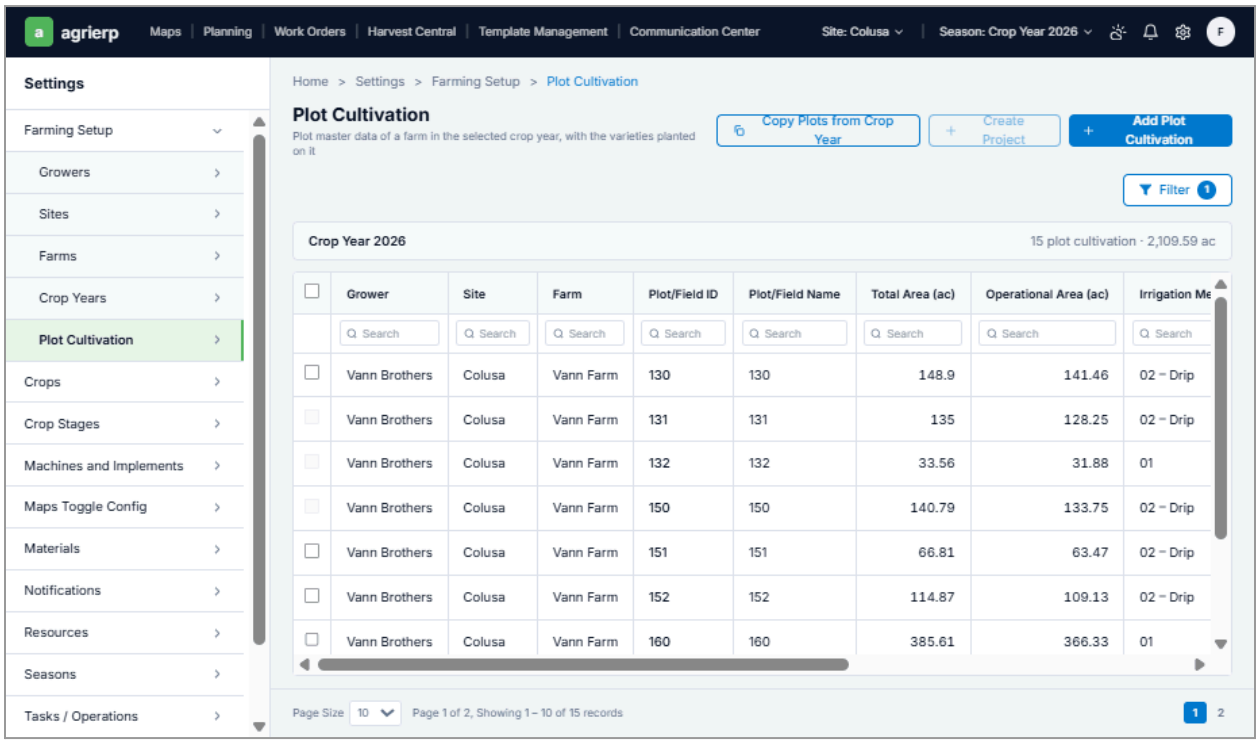
R1 · Farms



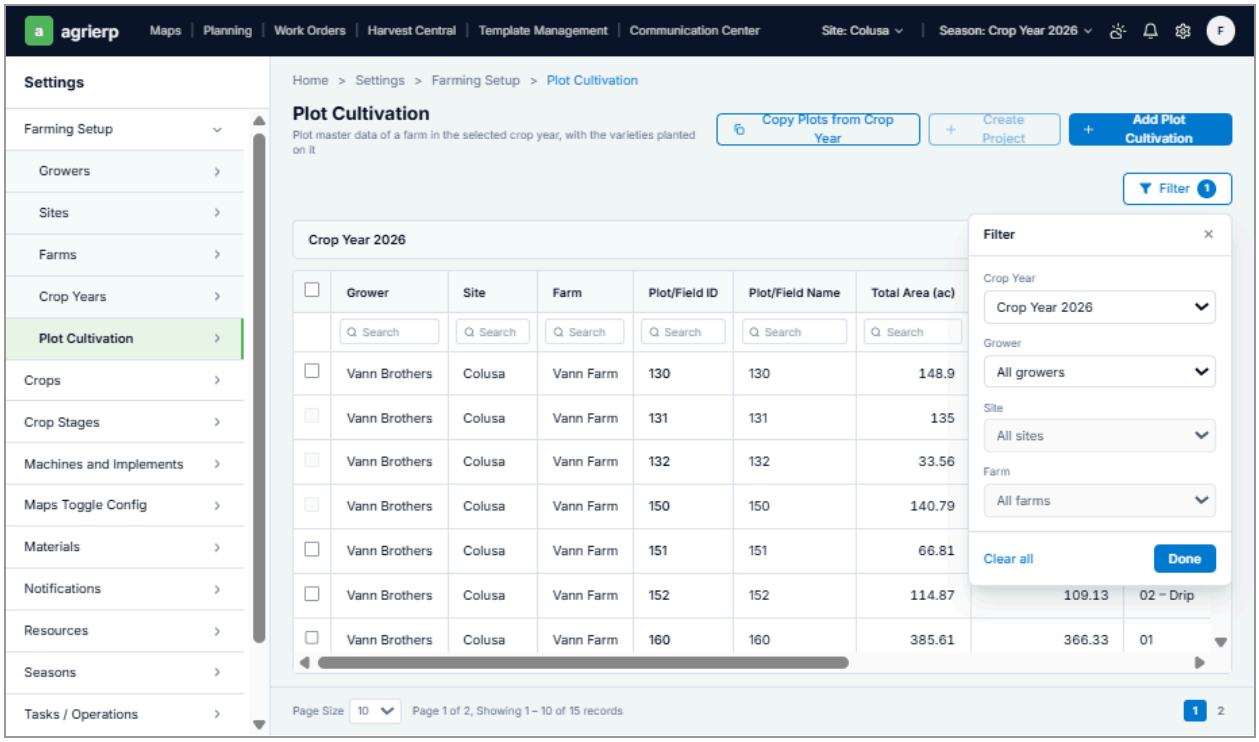
R1 · Crop Years (create empty)



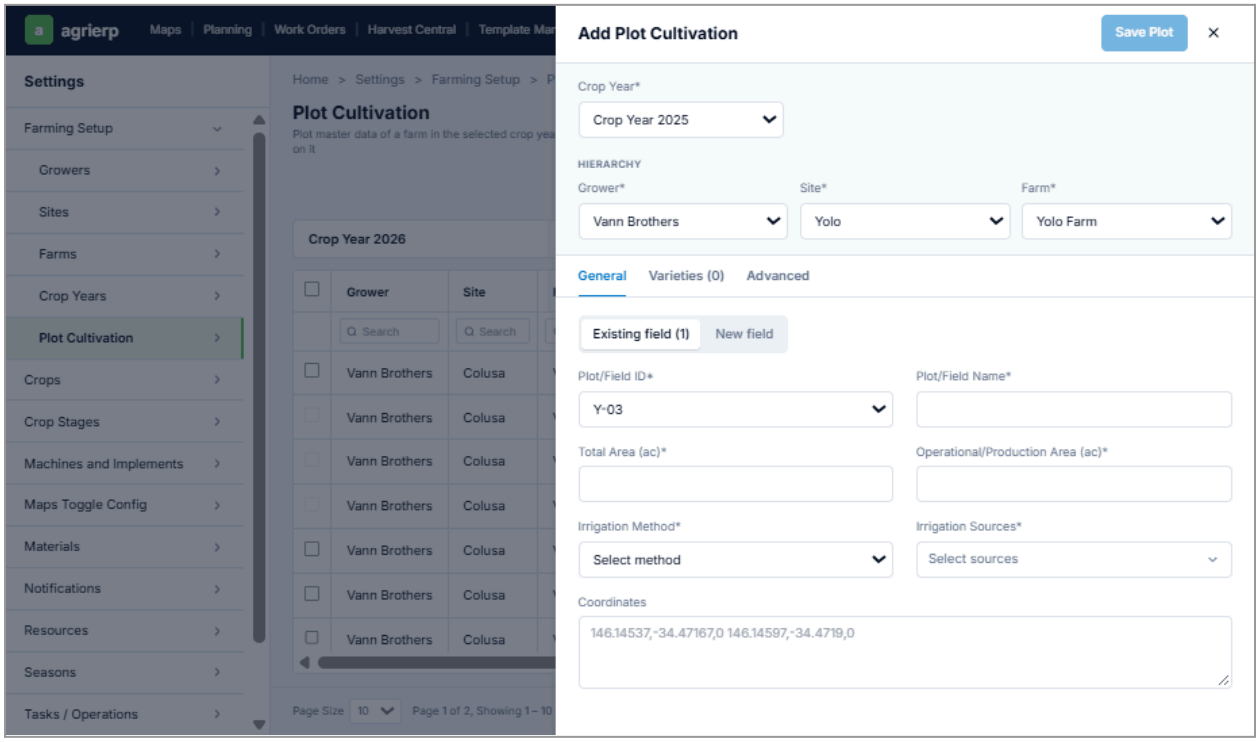
R2 · Plot Cultivation



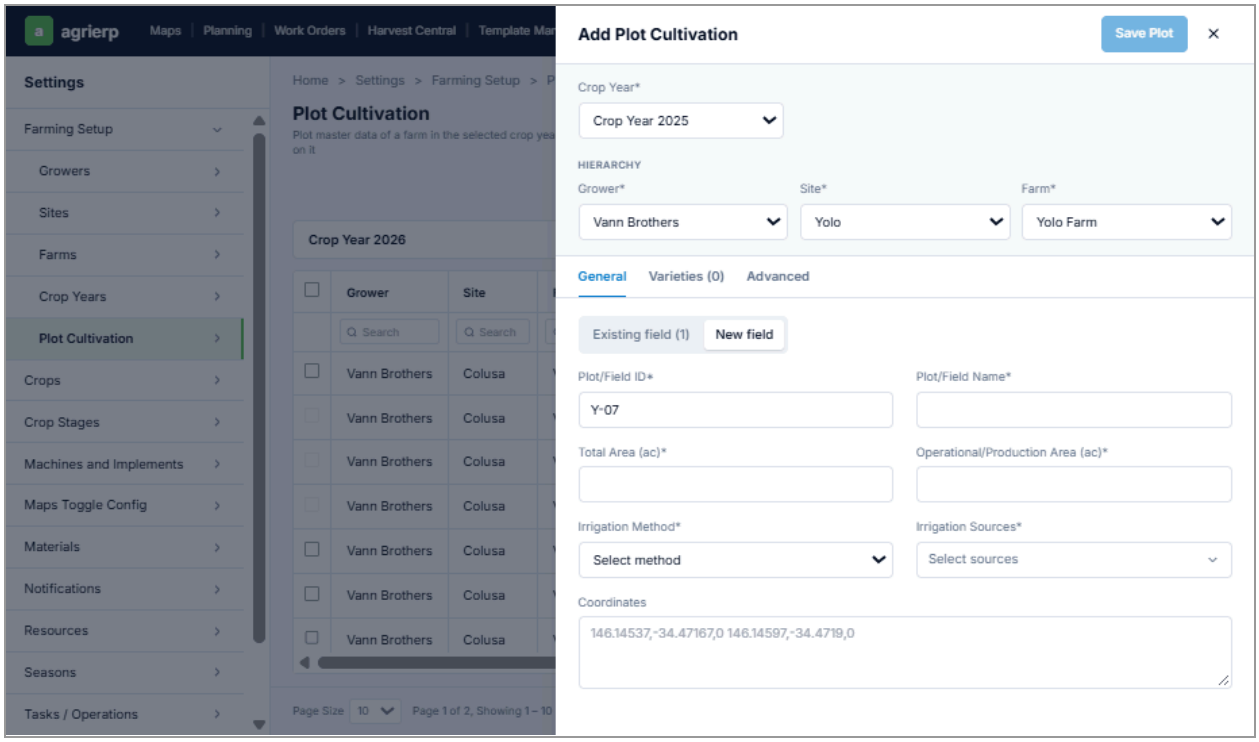
R2 · Filter (Crop Year, Grower, Site, Farm)



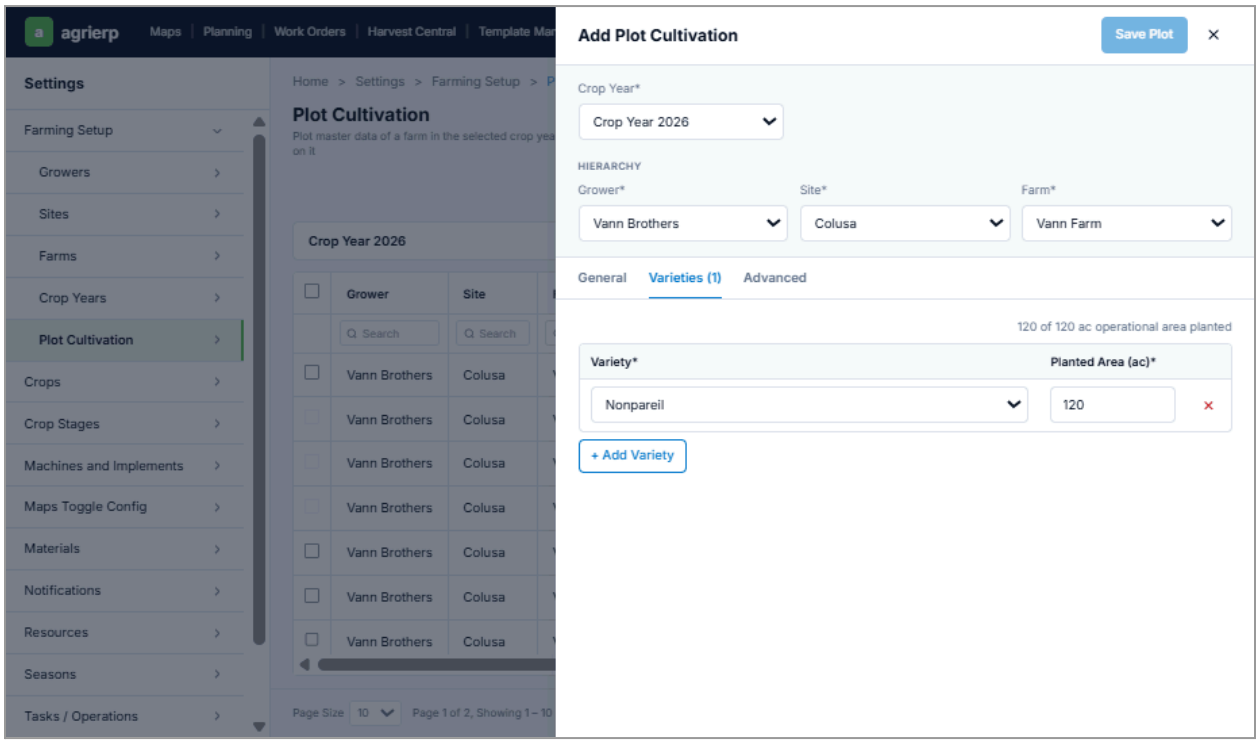
R2 · Add Field Cultivation › Existing Field — Field ID selected from the Field Master and read only; details entered for this crop year



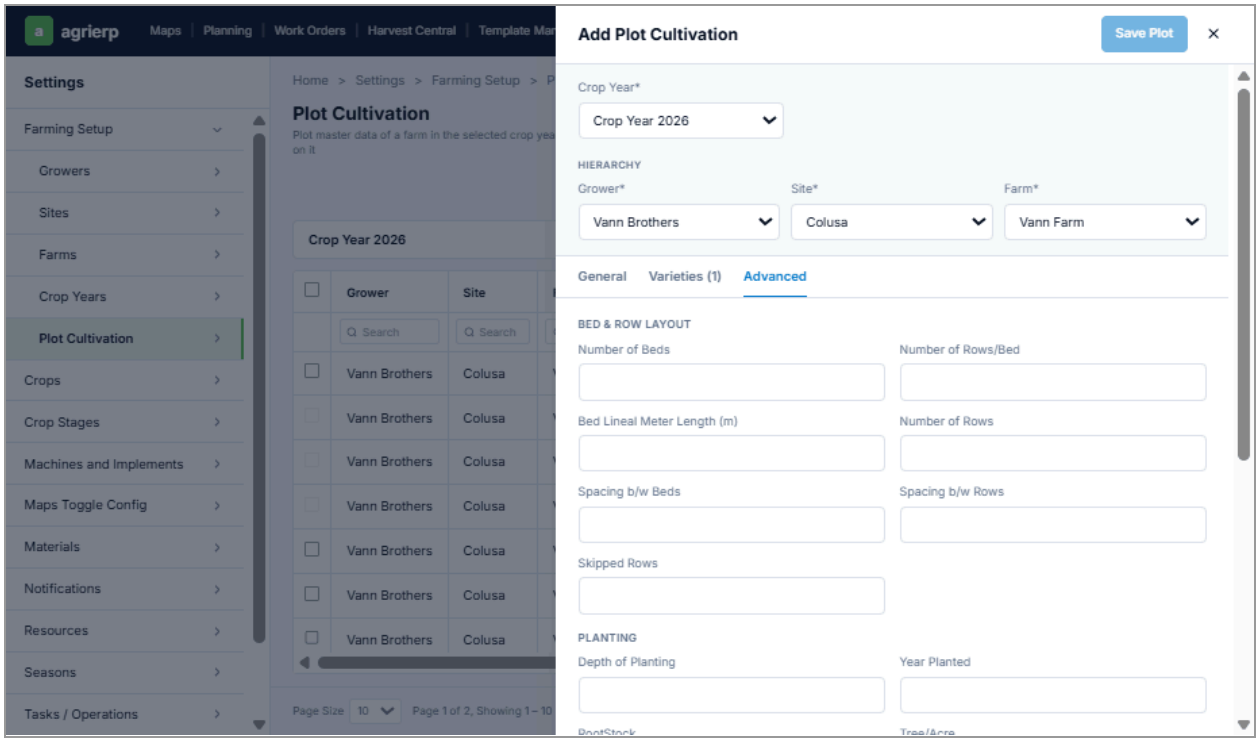
R2 · Add Field Cultivation › New Field — Field Master created only when the Field ID does not exist yet



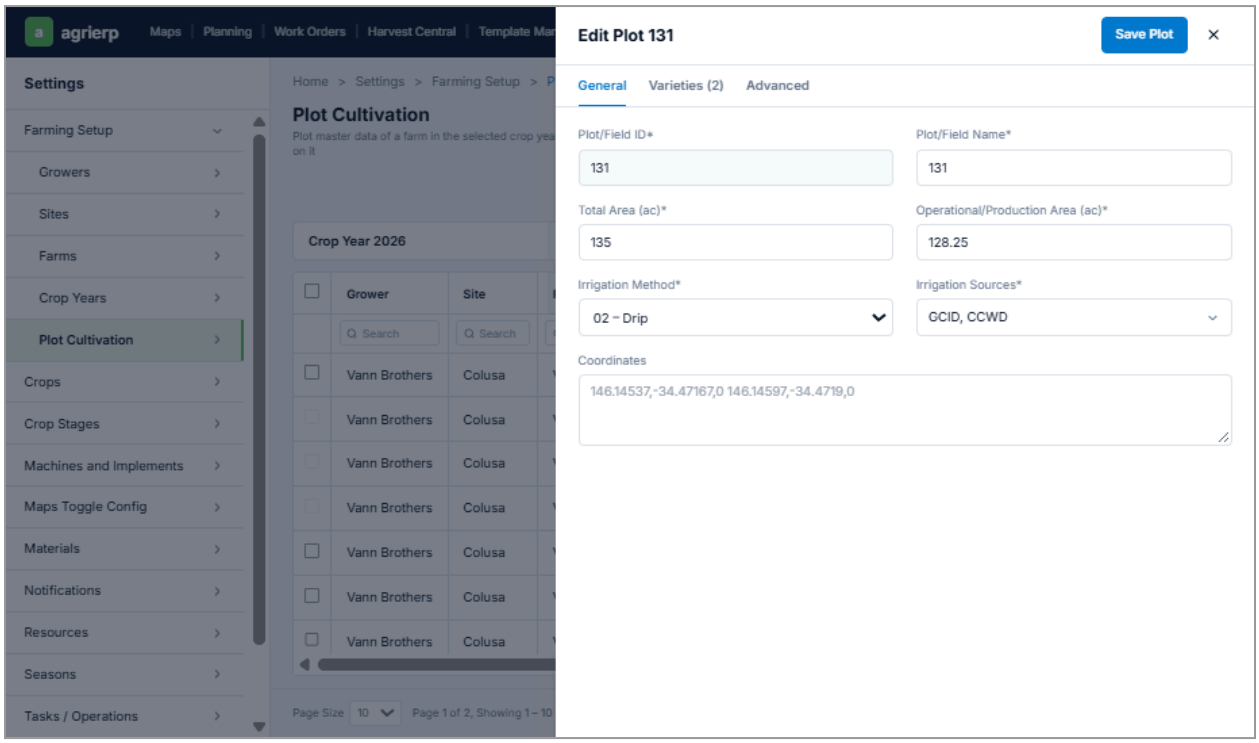
R2 · Add Plot Cultivation › Varieties



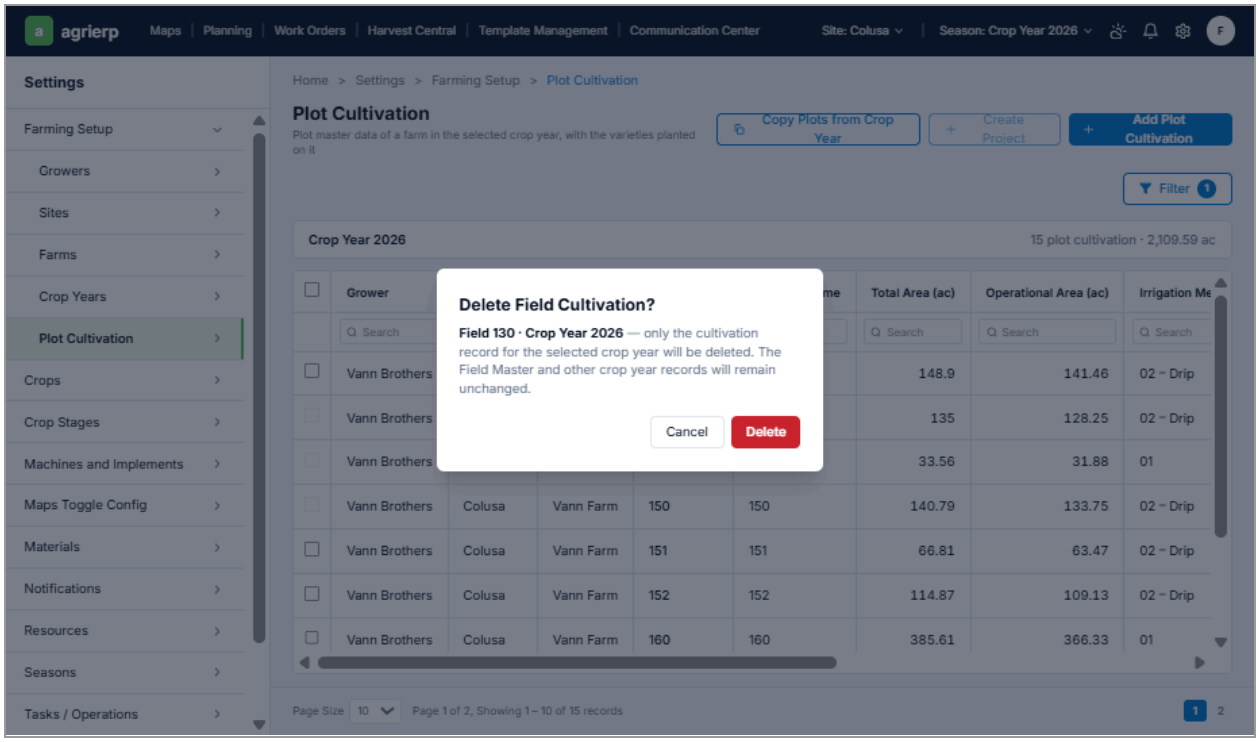
R2 · Add Plot Cultivation › Advanced



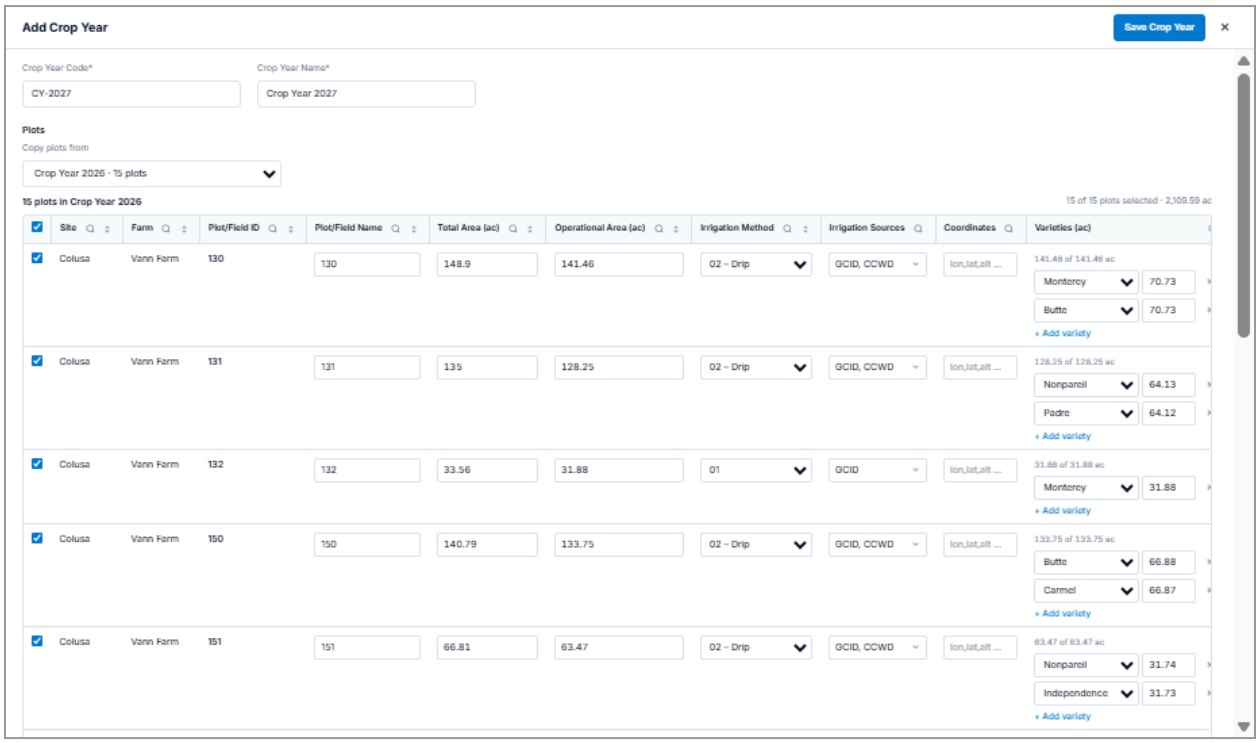
R2 · Edit Plot › General



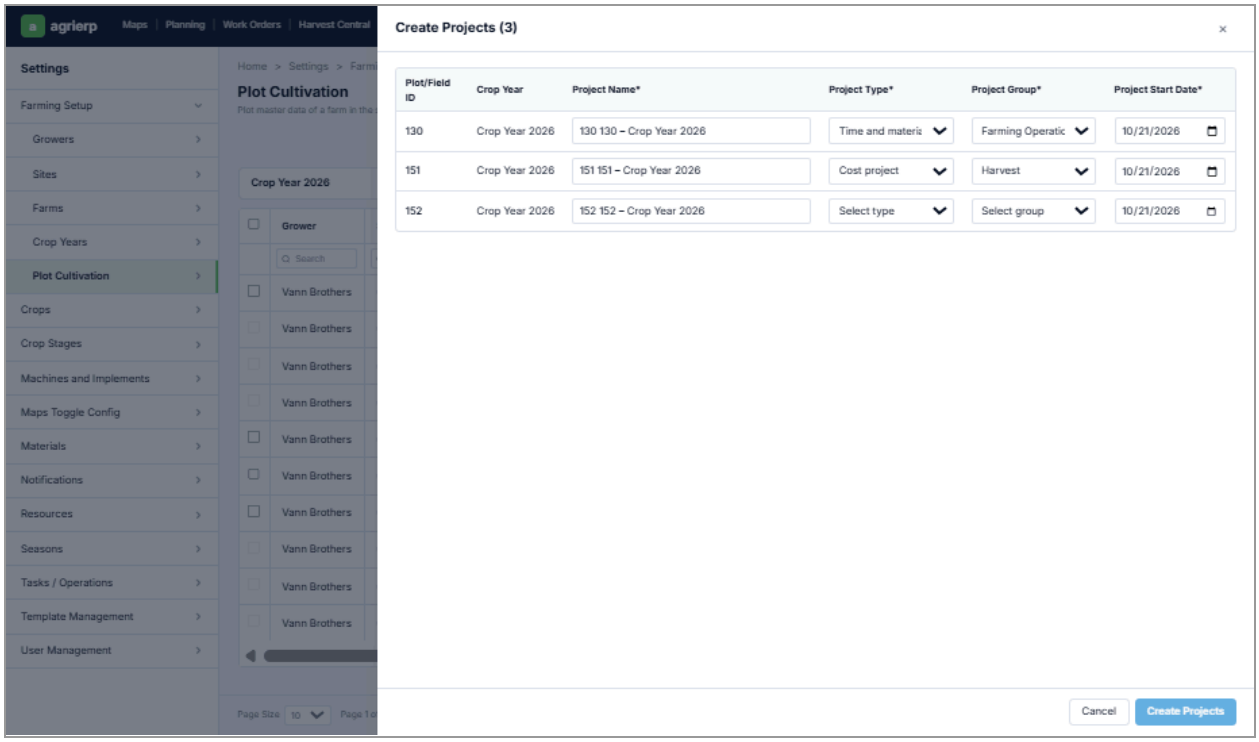
R2 · Delete Field Cultivation — Field Master and other crop years unchanged



R2 · Create Crop Year with Plot Copy



R3 · Create Projects — one row per selected plot


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Maps
Planning
Work Orders
Harvest Central
Settings
Farming Setup
Growers
Sites
Farms
Crop Years
Plot Cultivation
Crops
Crop Stages
Machines and Implements
Maps Toggle Config
Materials
Notifications
Resources
Seasons
Tasks / Operations
Template Management
User Management
Home > Settings > Farm
Plot Cultivation
Plot master data of a farm in this
Crop Year 2026
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Page 1 of
Create Projects (3)
Plot/Field ID
Crop Year
Project Name*
Project Type*
Project Group*
Project Start Date*
130
Crop Year 2026
130 130 – Crop Year 2026
Time and materia
Farming Operatic
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151
Crop Year 2026
151 151 – Crop Year 2026
Cost project
Harvest
10/21/2026
152
Crop Year 2026
152 152 – Crop Year 2026
Select type
Select group
10/21/2026
Cancel
Create Projects

App User

Access is feature-based: the Farming Setup feature (pages and actions, including D365 Project Creation and Project Update) is granted to the Farm Admin and Farm Operations Manager roles. The same permission is checked in the UI and in the backend/API.

App Access	Roles	Description
Farm Web App	Farm Admin	Views, adds, edits and deletes Growers, Sites, Farms, Crop Years and Field Cultivation records; copies; creates and updates D365 projects.
Farm Web App	Farm Operations Manager	Same access as Farm Admin: views, adds, edits and deletes Growers, Sites, Farms, Crop Years and Field Cultivation records; copies; creates and updates D365 projects.
Farm Web App	All other roles	No access — Farming Setup pages and actions are not shown, and direct calls are refused by the server.
D365 F&SCM	Integration user	Service account used by AgriERP to create and update projects; sets AgriProject = Yes.
Farm Mobile App	Machine Operator	Out of scope — consumes plot, variety and project data read-only.

Business Use Cases

One business case per real need. Screen-level behaviour and exceptions are listed once, in Detail Cases at the end of this document.

R1 — Hierarchy & Crop Year Setup

BC-01 · See the existing growers, sites and farms in AgriERP

- Need: the hierarchy already exists in the database but there is no screen for it; changes depend on scripts.
- Who: Farm Admin, Farm Operations Manager.
- Outcome: Growers, Sites and Farms pages list every existing record with its current ID and name, searchable by any column, with counts that open the child list.

BC-02 · Onboard a new grower and their land

- Need: a newly signed grower and their ranches and land blocks must exist before anything can be farmed or costed.
- Who: Farm Admin, Farm Operations Manager.
- Outcome: Grower, then Site under it, then Farm under the site are added by entering names; IDs are assigned using the existing numbering scheme.

BC-03 · Keep the hierarchy clean

- Need: names get corrected and unused locations retired without breaking history.

- Outcome: names can be edited (IDs and parents never change); a record can be deleted only when nothing is linked below it.

BC-04 · Open a new season

- Need: every cultivation record and project belongs to a crop year.
- Who: Farm Admin, Farm Operations Manager.
- Outcome: an empty Crop Year is created with the company's Code and Name. Filling it with last year's fields is done in R2 (BC-08).

R2 — Field Master & Crop-Year-Wise Cultivation

BC-05 · Keep one permanent Field Master per physical field

- Need: reports must follow the same field across seasons.
- Outcome: each field exists once in the Field Master under its Farm with a permanent, unique Field ID; existing fields are shown. When a season is recorded the user selects the existing field; a new Field Master is created only for a genuinely new field. All crop years of a field can be opened from it.

BC-06 · Record how a field is farmed this season

- Need: acres, irrigation, location and varieties change from year to year and drive work orders, harvest and costing.
- Who: Farm Admin, Farm Operations Manager.
- Outcome: a Field Cultivation record is added for the selected Field ID + Crop Year with Total and Operational Area, Irrigation Method and Sources, Coordinates, Varieties with planted acres , Planted Year and other Advanced details. Planted acres can never exceed the Operational Area, and Operational Area never exceeds Total Area.

BC-07 · Correct or remove a season's record

- Need: mistakes in a season's entry must be fixed without touching other seasons.
- Outcome: the cultivation record can be viewed, edited or deleted for that crop year only. The field and its other years stay unchanged.

BC-08 · Roll over to the next season

- Need: avoid re-keying hundreds of fields every year.
- Who: Farm Admin, Farm Operations Manager.
- Outcome: Create Crop Year with Plot Copy (or copy into an existing crop year) reuses the existing Field Master and creates new Field Cultivation records with the previous year's details as starting values. Acres, irrigation, coordinates, varieties and other crop-year details can be changed before saving; fields not farmed are unselected. No Project No. is copied, no duplicate Field ID + Crop Year is created and the previous year stays unchanged.

BC-09 · Protect seasons that are already being costed

- Need: some cultivation records are already linked to D365 projects today.
- Outcome: any cultivation record with a linked project — existing or created later in R3 — cannot be deleted and is read only except adding a new variety, until R4.

R3 — D365 Project Creation

BC-10 · Give every season's field a project before costs are booked

- Need: work orders and harvest costs post only to a D365 project.
- Who: Farm Admin, Farm Operations Manager.
- Outcome: one or many cultivation records without a project are selected and a project is created for each, with its own Name, Type, Group and Start Date. D365 returns the Project No.; AgriProject = Yes is set by the integration and is not visible or editable in AgriERP.

BC-11 · Recover when D365 refuses a project

- Outcome: the record shows the D365 message; the user corrects the value and creates again. Only one project can exist per Field + Crop Year.

R4 — D365 Project Update

BC-12 · Keep the D365 project in step with the field

- Need: per-acre costing must use the acres actually farmed.
- Who: Farm Admin, Farm Operations Manager.
- Outcome: Once a D365 project is linked to a Field Cultivation record, any updates to its D365-mapped fields must be synchronized automatically when the user saves the changes in AgriERP. Updates to fields that are not mapped to D365 remain in AgriERP only. Save triggers the sync automatically for the project linked to that Field + Crop Year only. If D365 rejects the update, AgriERP keeps the saved change, shows Update failed and offers Retry without creating a duplicate project. The exact mapped field list depends on the D365 mapping review; automatic sync on Save is confirmed.

Validation Rules

General (all releases)

- Farming Setup is a permissioned feature granted to Farm Admin and Farm Operations Manager. Both roles can view, add, edit, delete and copy Growers, Sites, Farms, Crop Years and Field Cultivation records, and create and update D365 projects, subject to the rules below. Users with any other role cannot open Farming Setup pages or perform its actions; the same check is enforced in the UI and in the backend/API.
- Field ID: the permanent identity of a physical field, required, max 50, trimmed. Never changeable and never re-created for a new crop year. Whether it is unique per Farm or company-wide is pending (see Open Questions).

- Field Master ID + Crop Year is unique: a Field Master has only one cultivation record per crop year. Creating a cultivation record never creates another Field Master for an existing field.
- Total Area, Operational Area, Irrigation Method and Sources, Coordinates, Varieties, Planted Year and the other cultivation attributes belong to the Field Cultivation record; editing one crop year never changes another.
- Plot/Field Name: required, max 50.
- Total Area and Operational/Production Area: required, numeric > 0, max 2 decimals. Operational Area ≤ Total Area.
- Irrigation Method: required, single select. Irrigation Sources: required, one or more (multi-select dropdown).
- Varieties: each row needs a variety and planted area > 0; a variety appears once per plot; sum of planted areas ≤ Operational Area
- Advanced fields are optional; numeric fields accept digits (and decimals where marked), max 50.
- A cultivation record belongs to one Field and one Crop Year; the Field, its Farm and the Crop Year cannot change after save.
- Every add, update, delete, copy and project call writes an audit entry.

R1 — Hierarchy & Crop Year Setup

- Existing Grower, Site and Farm records from the database are shown with their current IDs. New records get an ID from the existing numbering scheme; the ID is read only and never re-used.
- Names are required (max 50) and unique within the parent; only the name can be edited.
- A Grower, Site or Farm can be deleted only when nothing is linked below it.
- Hierarchy pickers fill top-down; children reset when a parent changes; a level with one option is auto-selected.
- Crop Year Code: entered by the user (suggested as CY-YYYY), required, unique, max 20, locked after save. Crop Year Name: required, unique, max 50.
- R1 creates empty crop years only. A crop year with cultivation records cannot be deleted.

R2 — Field Master & Crop-Year-Wise Cultivation

- Add Field Cultivation offers two options. Existing Field: the user selects a Field Master; its Field ID is read only and a cultivation record is created for the selected crop year. No details are copied from earlier crop years — the user enters them for this year. New Field: the user enters a Field ID that does not exist yet; a Field Master is created only in this case, together with the cultivation record.
- Existing Field lists only fields that have no cultivation record in the selected crop year.
- Cultivation records can be added, viewed, updated (except Plot/Field ID, Crop Year and hierarchy) and deleted for one crop year at a time.

- Deleting a cultivation record removes only that crop year's record and its varieties; the Field Master and other crop year records stay unchanged. Cultivation records linked to a project cannot be deleted.
- Any cultivation record already linked to a project in the database or D365 cannot be deleted and is read only except adding a new variety and its planted area.
- Add Crop Year copy: Operational Area defaults to the source value; when it equals Total Area it follows Total Area as the user types. Grower, Site, Farm and Plot/Field ID are copied as-is and locked. Copy reuses the existing Field Master, creates new Field Cultivation records with the previous year's values as starting values, never carries a Project No., never creates a duplicate Field ID + Crop Year and leaves the source year unchanged.
- Copy into an existing year skips plots that already exist there (same Field Master ID + Crop Year); they are shown as "Already in …" and cannot be selected.
- Save is blocked while any selected copy row breaks a rule; the row's cells are outlined red.

R3 — D365 Project Creation

- Project creation is triggered only when an authorized user selects one or more cultivation records without a linked project, enters Project Name, Project Type, Project Group and Project Start Date, and clicks Create Project. Creating or saving a cultivation record never creates a D365 project.
- D365 creates a separate project per selected Field Master + Crop Year record and returns its Project No., which is saved on that record. A second project is never created for the same Field Master + Crop Year.
- Only selected plots without a Project No. are sent; a plot with a project cannot be selected.
- Project Name, Project Type, Project Group and Project Start Date are required for every plot. Each plot has its own values.
- Project Name defaults to "{Plot ID} {Plot Name} – {Crop Year}" (max 60); Project Start Date defaults to the creation date and can be changed.
- AgriProject is set to Yes by the integration on every project created from AgriERP. It is not shown on any AgriERP screen and cannot be changed by a user.
- One project per Plot + Crop Year — a second create request for the same pair returns the existing Project No.
- While a create is in progress the row shows Creating… and cannot be edited or deleted.
- After creation the same lock as existing linked projects applies (until R4).

R4 — D365 Project Update

- A cultivation record with a project — created in R3 or linked before AgriERP — can be updated except Plot/Field ID, Crop Year and hierarchy. Delete stays blocked.
- Once a D365 project is linked to a Field Cultivation record, any updates to its D365-mapped fields must be synchronized automatically when the user saves the changes in AgriERP. Updates to fields that are not mapped to D365 remain in AgriERP only.

- No linked project → Save stores the changes in AgriERP only. Linked project and only AgriERP-only attributes changed (varieties, advanced) → saved without any D365 call.
- Sync is triggered automatically on Save; there is no separate sync action. Changes go only to the project linked to that Field + Crop Year, never to another crop year's project.
- Only fields with a D365 mapping are sent. Project Name, Total Area, Operational Area and Irrigation Method stay To confirm in the mapping table until their exact D365 targets are verified; the review decides which fields are synced, not whether sync on Save is required. AgriProject stays Yes and is never sent as a user value.
- If D365 rejects the update the saved AgriERP change is kept, the record shows Update failed with the D365 message and a Retry action. Retry re-sends the latest saved values to the same project and never creates a duplicate project.

Messages

When	Message
Plot saved	Plot {ID} added in {Crop Year} / Plot {ID} updated in {Crop Year}
Saved — project linked, mapped field changed (R4)	Plot {ID} saved · updating Project {Project No.} in D365…
Project update succeeded (R4)	Project {Project No.} updated in D365
Project update failed (R4)	Project {Project No.} not updated — D365 rejected the change. Saved in AgriERP; use Retry.
Duplicate Field Master	Field ID already exists. Please select the existing field.
Duplicate Field Cultivation	A cultivation record already exists for Field {Field ID} in Crop Year {Crop Year}.
Operational > Total	Can't be more than the total area / More than total area
Varieties > Operational	{X} of {Y} ac — more than operational area
Delete Field Cultivation	Only the cultivation record for the selected crop year will be deleted. The Field Master and other crop year records will remain unchanged.
Delete hierarchy record	{ID} has nothing linked to it and will be removed. This can't be undone.
Delete blocked	Can't delete — Project {Project No.} already created / Linked to {n} plots
Crop year created	{Crop Year} created with {n} plots
Copy done	{n} plots copied to {Crop Year}
Project create started / done	Creating {n} projects in D365… / Project {No.} created for plot {ID}
Project create failed	D365 error: {message}. Fix the plot and create again.

Future Considerations:

- Plot Hierarchy customisation — rename levels, switch Site/Farm off, add custom levels.
- Plot Fields customisation — show/hide fields and make optional fields required per customer.
- Bulk import from the Excel/CSV template and KML coordinates; map view of plots.
- Pull Project Type and Project Group lists live from D365 instead of configured values.

Technical Specification

The model below describes what the feature needs logically. It must be mapped to the existing database schema; new tables or ID formats are added only where the existing schema has no equivalent.

Logical Data Model

Entity	Holds	Existing schema
Grower / Site / Farm	id, parent, code, name	Existing tables; reuse current IDs and numbering.
Crop Year	id, code, name	Existing table if present.
Field Master	id, farmId, fieldId (permanent Field ID)	One row per physical field; never re-created per crop year. Only attributes confirmed as permanent belong here; none beyond Field ID and Farm are confirmed yet. Field ID uniqueness scope (per Farm or company-wide) pending.
Field Cultivation	fieldMasterId, cropYearId, name, totalArea, opArea, irrigationMethod, irrigationSources, coordinates, plantedYear, other cultivation attributes, copiedFromId	Unique (fieldMasterId, cropYearId).
Cultivation Variety	cultivationId, variety, plantedArea	Σ planted Area ≤ opArea.
Cultivation Project	fieldCultivationId, projectNo, name, type, group, startDate, status, error, lastSyncedAt	Linked to the Field Cultivation record, never to the Field Master. AgriProject lives in D365 only.
Audit	entity, action, before, after, user, time	Use the existing audit mechanism.

IDs

- Grower, Site and Farm: keep the existing IDs; new records use the existing sequence/format.
- Crop Year Code and Plot/Field ID: entered by the user and validated for uniqueness on the server.

D365 F&SCM Integration

Automatic synchronization on Save for project-linked cultivation records is confirmed. Still to confirm: the exact D365 target fields for Project Name, Total Area, Operational Area and Irrigation Method — these stay To confirm until the D365 consultant and integration owner verify them. The technical integration approach is also to be validated with the D365 integration team.

AgriERP field	D365 Project field	Status	Rule
Project Name	Project name	Confirmed	Set on create (R3).
Project Type	Project type	Confirmed	Set on create.
Project Group	Project group	Confirmed	Set on create; must exist in the legal entity.

AgriERP field	D365 Project field	Status	Rule
Project Start Date	Project start date	Confirmed	Set on create.
— (system)	AgriProject	Confirmed	Always Yes; set by the integration; never changed.
Field Master + Crop Year	Project ID (returned)	Confirmed	One project per Field Cultivation record; stored on that record.
Project Name (change)	Project name	To confirm	R4 — synced automatically on Save once the D365 target is verified.
Total / Operational Area	Growing plot master areas	To confirm	R4 — synced automatically on Save once the D365 target is verified.
Irrigation Method	Growing plot master	To confirm	R4 — synced automatically on Save once the D365 target is verified.
Varieties, Advanced	—	Not synced	AgriERP only.

Project Status Flow

No project → Creating… → Created (Project No. shown) or Failed (D365 message). R4, on Save of a project-linked record with a mapped change: Created → Updating… → Synced (last synced time) or Update failed (D365 message, saved AgriERP change kept) → Retry → Updating… to the same project. Saves without a mapped change make no D365 call and leave the status unchanged. Calls are idempotent per cultivation record and use the integration service account; retry policy to be agreed with engineering.

Farming Setup - Data Dictionary

Section	Field Name	Required	Data Type	Max Length	Release	Other Info
Plot hierarchy	Crop Year	yes	Dropdown	50	R2	Lookup: Crop Years
	Grower	yes	Dropdown	50	R2	Lookup: Growers
	Site	yes	Dropdown	50	R2	Filtered by Grower
	Farm	yes	Dropdown	50	R2	Filtered by Site
Field Master	Field ID	yes	Text / Select	50	R2	Permanent; read only for an existing Field Master; uniqueness scope pending
Field Cultivation · General	Plot/Field Name	yes	Text	50	R2	Per Field ID + Crop Year
	Total Area (ac)	yes	Decimal(2)	25	R2	> 0
	Operational/Production Area (ac)	yes	Decimal(2)	25	R2	> 0, ≤ Total Area
	Irrigation Method	yes	Dropdown	50	R2	
	Irrigation Sources	yes	Multi-select dropdown	50	R2	At least one
	Coordinates	no	Text (lon,lat,alt list)	20000	R2	From KML
Varieties	Variety	yes	Dropdown	50	R2	Unique per plot
	Planted Area (ac)	yes	Decimal(2)	25	R2	> 0; Σ ≤ Operational Area
Advanced · Bed & Row	Number of Beds	no	Integer	50	R2	
	Number of Rows/Bed	no	Integer	50	R2	
	Bed Lineal Meter Length (m)	no	Decimal	50	R2	
	Number of Rows	no	Integer	50	R2	
	Spacing b/w Beds	no	Decimal	50	R2	
	Spacing b/w Rows	no	Decimal	50	R2	
	Skipped Rows	no	Integer	50	R2	
Advanced · Planting	Depth of Planting	no	Decimal	50	R2	
	Year Planted	no	Integer	4	R2	YYYY; per crop year
	RootStock	no	Text	50	R2	
	Tree/Acre	no	Integer	50	R2	
	Plants/Acre	no	Integer	50	R2	

Section	Field Name	Required	Data Type	Max Length	Release	Other Info
Advanced · Land & Records	FSA Reported Area	no	Decimal	50	R2	ac
	Lease Land Area	no	Decimal	50	R2	ac
	APN#	no	Text	50	R2	
	Old Field#	no	Text	50	R2	
	LandOwner or Company Name	no	Text	50	R2	
	Range	no	Text	50	R2	
	Other	no	Text	50	R2	
Crop Year	Crop Year Code	yes	Text	20	R1	User entered (suggested CY-YYYY); unique; locked after save
	Crop Year Name	yes	Text	50	R1	Unique
	Copy plots from	no	Dropdown	—	R2	Source crop year or “Don’t copy”
Hierarchy pages	Grower / Site / Farm ID	system	Text	—	R1	Existing ID / existing numbering scheme; read only
	Grower / Site / Farm Name	yes	Text	50	R1	Unique within parent
Project	Project Name	yes	Text	60	R3	Default “{ID} {Name} – {Crop Year}”
	Project Type	yes	Dropdown	50	R3	Per plot
	Project Group	yes	Dropdown	50	R3	Per plot
	Project Start Date	yes	Date	—	R3	Default creation date; editable
	AgriProject	system	Yes/No	—	R3	D365 only; always Yes; not on UI
	Project No.	system	Text	20	R3	Returned by D365; read only
	Project Status	system	Enum	—	R3	None / Creating / Created / Failed / Updating / Synced / Update failed
	Last Synced At	system	DateTime	—	R4	

Detail Test Cases

Each line reads: situation → expected behaviour. Together they cover every normal and exception path of the business cases above.

R1 — Hierarchy & Crop Year Setup

Existing data

- Growers, Sites or Farms page is opened → all records already in the database are listed with their current IDs and names.
- Search is typed under any column → the list narrows; several column searches combine; Clear resets them.
- A count link (e.g. “3 farms”) is clicked → the child page opens filtered to that parent.

Add

- Add Grower / Site / Farm is opened → the ID field is read only and filled from the existing numbering scheme on save.
- Two users add a record at the same time → each gets a different ID.
- Name is empty → save blocked. Same name under the same parent → save blocked with a message.
- Add Site without a Grower, or Add Farm without a Site → the child picker stays disabled.
- The parent has only one child option → it is selected automatically; changing the parent clears the child.

Edit and delete

- Edit is opened → only the name can change; ID and parent are read only.
- Delete is clicked on a record with children or fields linked → disabled with “Linked to n …”.
- Delete is confirmed on a record with nothing linked → removed; its ID is never given to a new record.

Crop Year

- Add Crop Year is opened in R1 → Code (suggested, editable) and Name only; no copy option.
- Code or Name empty, or already used → save blocked.
- Existing crop year is edited → Name editable, Code read only.
- Delete on a crop year with cultivation records → disabled with “Linked to n plots”.

R2 — Field Master & Crop-Year-Wise Cultivation

Field Master

- Add Plot Cultivation on a farm with fields not yet cultivated in this crop year → choose Existing field (list of those fields) or New field.
- An existing field is selected → its Field ID is shown read only; the cultivation details are empty and entered for this crop year; no new Field Master is created.
- New Field with a Field ID that already exists → blocked: “Field ID already exists. Please select the existing field.”
- A field that already has a record in the selected crop year → not offered under Existing Field; a direct attempt is blocked: “A cultivation record already exists for Field {Field ID} in Crop Year {Crop Year}.”
- Plot/Field ID on an existing field → always read only.
- A cultivation record is added for a field that already has one in that crop year → blocked.
- View → Crop Years tab on Field 130 → every crop year of the field with its own acres, varieties and project; each opens independently.
- The project shown on a row → always the project of that Field ID + Crop Year.

Find

- Plot Cultivation is opened → the current crop year is filtered; the Filter badge shows the number of active filters.
- Grower is not chosen in Filter → Site and Farm stay disabled. Clear all removes every filter including Crop Year.
- Filter, column search and paging are combined → results and totals stay correct; a new search returns to page 1. No match → empty-state message.

Add and edit a cultivation record

- Filter is set to a farm → Add opens with Crop Year, Grower, Site and Farm filled; details stay disabled until all four are chosen.
- A required field is missing → shown in red; its tab gets a red dot; on save the panel switches to the first tab with an error.
- Total Area 0, empty or not a number → blocked. More than two decimals → limited to two.
- Operational Area equal to Total Area → accepted; larger → “Can’t be more than the total area”.
- No irrigation source → blocked; several sources can be picked.
- Add Variety → the new row is pre-filled with the operational acres not yet planted; a variety already used is not offered again.
- Variety acres exceed Operational Area → red total, red dot on Varieties, save blocked.
- A variety row has no acres → blocked. A variety is removed → the total updates immediately.

- Advanced left empty → saves. Letters in a numeric Advanced field → not accepted; Year Planted → 4 digits.
- A cultivation record is edited in one crop year → the same field in other crop years is unchanged.
- View is clicked → path, General, Varieties, Advanced and Project shown read only, with Edit and Delete available.

Delete

- Delete on a cultivation record without a project → “Only the cultivation record for the selected crop year will be deleted. The Field Master and other crop year records will remain unchanged.” On confirm only that year’s record and its varieties are removed.
- Delete is cancelled → nothing changes.
- Delete on a field that has cultivation records in any year → not allowed.

Records already linked to a project

- A cultivation record is linked to a project that existed before AgriERP → Delete disabled with “Project … already created”.
- Edit on such a record → General and Advanced read only; a new variety and its acres can be added; existing varieties cannot be changed or removed.

Create Crop Year with Plot Copy

- A source crop year is chosen → all its cultivation records are listed with Site, Farm and Plot/Field ID read only and Name, Total Area, Operational Area, Irrigation Method, Irrigation Sources, Coordinates and Varieties editable.
- Operational Area equals Total Area and Total Area is changed → Operational Area follows. Operational Area above Total Area → red; save blocked.
- Variety acres above Operational Area on a row → red; save blocked.
- Rows are unselected or the header checkbox is used → only selected fields get a cultivation record.
- Save → new Field Cultivation records on the existing Field Masters; no new Field Master; no Project No.; no duplicate Field ID + Crop Year; the source year unchanged.
- “Don’t copy — start with no plots” → empty crop year.
- Copy into an existing crop year → fields already cultivated there are marked “Already in …” and cannot be selected; if nothing is new, Copy is disabled.

R3 — D365 Project Creation

- No record is ticked → Create Project disabled.
- A record already linked to a project (existing or new) → its checkbox is disabled.
- Create Project (n) is opened → one row per record with Name pre-filled and Start Date = today.
- Different Type, Group and Start Date per row → each project created with its own values.

- Any row has an empty field → red; nothing is sent.
- Start Date in the past → project created with that date.
- Rows show Creating… → the record cannot be edited or deleted until the result returns.
- D365 returns the Project No. → shown on the row; the record is locked as in R2.
- D365 rejects (e.g. Project Group not in the legal entity) → row shows Failed with the message; the record stays editable and can be created again.
- The same record is sent twice → only one project exists for that Field + Crop Year.
- The project is opened in D365 → AgriProject = Yes. No AgriERP screen shows an AgriProject field.
- A new cultivation record is created or saved → no D365 project is created; the record stays without a Project No. until a user clicks Create Project.

R4 — D365 Project Update

- Edit on a record with a project (created in R3 or linked before) → everything except Plot/Field ID, Crop Year and hierarchy is editable, including existing varieties.
- A project-linked record is saved with a D365-mapped field changed → sync starts automatically; row shows Updating…, then Synced with the last synced time.
- A record without a linked project is saved → changes stored in AgriERP only.
- Field 130 has projects in 2026 and 2027 and the 2027 record is saved → only the 2027 project is updated.
- When Project Name, Total Area, Operational Area, or Irrigation Method changes, the system automatically synchronizes the changes on Save only where the respective D365 field mapping has been approved. The exact target fields remain subject to integration confirmation.
- Only non-mapped data changes (varieties, advanced) → saved in AgriERP only; no D365 call.
- Save with no change → no D365 call.
- D365 rejects the update → the saved AgriERP change is kept; row shows Update failed with the D365 message and Retry; Retry re-sends to the same project (no duplicate project) → Synced.
- Sync failed, then the user edits the record again and saves before retrying → the new values are saved in AgriERP and sync is attempted again on Save; Retry always sends the latest saved values, never the values from the failed attempt.
- Sync failed, then the user saves again with only AgriERP-only changes → no D365 call; status stays Update failed and Retry sends the latest saved mapped values.
- Delete on a record with a project → still not allowed.
- The project is checked after an update → AgriProject still Yes.